Sang Hak Lee

Marine Data Engineer · Vessel & weather data collection, datasets and quality

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Data engineer who **collects and cleans** AIS / V-PASS vessel data and marine-weather forecast data (NetCDF), and **builds datasets** on PostgreSQL data models I design. Embeds spec-based **data quality validation** and processing **metadata management** into pipelines; licensed weather forecaster with meteorology and ocean-environment certifications, bringing domain understanding of vessel, weather and route data.

EXPERIENCE

GeoSR (Geo System Research) · Data Engineer

Feb 2023 - Present

- **Vessel data collection & cleansing** — Cleaned TB-scale AIS trajectories for the Busan area (outliers, duplicates, time alignment) and joined route/anchorage data to build anchoring, loitering and route-deviation candidate datasets (GeoPandas, multiprocessing).
- **V-PASS dataset construction** — Streaming merge and filtering of vessel observations (data years 2023-2025) into regional/grid statistical datasets and density maps, joined with catch-grade data.
- **Weather & ocean data pipeline** — Automated NetCDF/CSV ingestion → spec validation → raster and current-vector conversion → PostgreSQL loading → GeoServer publishing (Python, Shell, Docker, air-gapped).
- **Data modeling** — Split loading for 11 monitoring/forecast products from 4 institutions into a **file catalog** (data type, year-month, monitoring/forecast, publication flag), **validation results** (per-file pass/fail and error items) and **value tables** (current lat/lon, speed, direction, depth), linked by file name.
- **Data quality validation** — Automated spec checks (Class, Variable, Level, Unit) with error records, exclusion and count reconciliation, service-point to model-grid matching, old vs. new product comparisons.
- **Metadata management** — Maintained processing/error history, input spec sheets and data specifications; traced source file → validation result → loaded values per file.
- **Systems & documentation** — Built forecast production servers (Oracle DB, Tomcat), scheduling, monitoring, incident response; requirements and test documents; conference poster (2024), workshop talk (2025).

Korea Meteorological Industry Promotion Agency · R&D Planning Intern

Jun - Sep 2022

- Researched weather technology and policy trends; wrote monthly newsletters and quarterly reports.

SELECTED PROJECTS

Marine Travel AI Agent · GeoSR internal R&D

2026

Managed public data consumed by an LLM agent — automated TourAPI and 8 marine-forecast feed collection (n8n), duplicate-ID, coordinate-range, missing-page and per-feed freshness checks, collection/issue timestamps, typed validation tools for LLM use (LangGraph, FastAPI).

AIS Trajectory Analytics

2024 - 2025

Activity-candidate detection from AIS tracks with route and anchorage polygons; KOSMEE autumn conference poster (2024) and workshop talk (2025).

Ocean Forecast GeoPipeline

2025.06 - 2026.04

NetCDF/CSV spec validation → spatial conversion → PostgreSQL loading → GeoServer publishing; code and validation docs public on GitHub.

SKILLS

Data	Python, SQL, PostgreSQL, Oracle, Pandas, GeoPandas
Formats	NetCDF, AIS, V-PASS, GeoTIFF, GeoServer, QGIS
Infra	Linux, Shell, Docker, scheduling & monitoring
Other	Fortran, Rust, LangGraph, FastAPI, Ollama (LLM agents)

CREDENTIALS · EDUCATION

Weather Forecaster License · Engineer Meteorology · Engineer Ocean Environment

B.S. Earth & Environmental Sciences, Jeonbuk National University (2016 - 2022)

Research Club Grand Prize, JBNU College of Natural Sciences (2021)

① Monitoring & forecast product loading pipeline

GeoSR · Jun 2025 - Apr 2026

Air-gapped pipeline that validates 11 monthly marine/weather monitoring and forecast products (NetCDF, CSV) from 4 institutions (KIOST, NIFS, KHOA, KOPRI), loads them into PostgreSQL and publishes GeoServer layers.

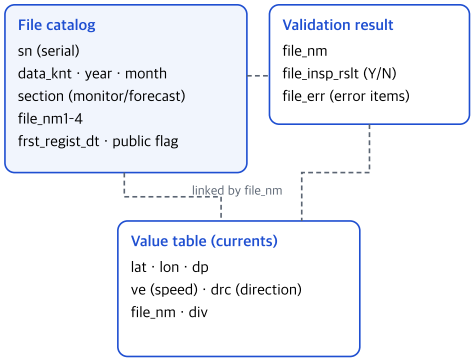
Role Data model design, spec validation, spatial conversion and ETL; specifications and test documents

Design Separated product **metadata** (data type, year-month, monitoring/forecast, registration time, publication flag), **validation results** and **values** into distinct tables, tracing source → validation → load per file

Validation Automatically compared Class, Variable, Level (depth) and Unit against the input spec; recorded pass/fail and error items per file

Processing File watching (Python, Shell, Docker), raster regridding/reprojection, speed and direction from U/V, upserts, transactions, reprocessing, scheduling

Results Item-level reports, GeoTIFF and current-vector CSV, GeoServer layers; code and validation docs public on GitHub



Loading structure (summary of actual tables)

If I redesigned it I would add a unique key on (data type, year-month, section), link all three tables with a file_id foreign key, and store speed/direction as numeric with a unit column for query performance and integrity.

② AIS vessel data cleansing & activity detection

GeoSR · 2024 - 2025

Combined historical Busan-area AIS with vessel metadata and route/anchorage polygons to detect and visualize anchoring, loitering and route-deviation candidates.

Role Trajectory preprocessing, spatial-rule and time-alignment detection, parallel visualization; poster and workshop talk

Cleansing Removed outliers and duplicates from TB-scale raw data, aligned tracks per vessel in time, derived speed, heading and displacement

Design Judged candidates with **readable rules** combining polygon membership and movement features so each decision can be reviewed

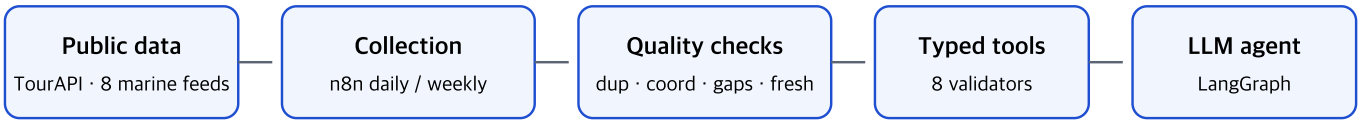
Processing GeoPandas spatial joins; multiprocessing for large map time-lapse generation

Results KOSMEE 2024 autumn conference poster, Jan 2025 workshop talk (10 pp.), public sample repo of 788K records / 1,070 vessels

③ Marine Travel AI Agent — public data for an LLM

GeoSR internal R&D · 2026

Collects and validates tourism, marine-forecast, weather and transport public data so an LLM agent can cite checkable evidence. Quality management of **external data an LLM consumes**, not training-dataset construction.



Role	Public-data integration, collection workflows, quality checks and typed validation tools, itinerary rule engine
Quality	Content-ID duplicates (0 found), coordinate range, missing/repeated pages, per-feed freshness; atomic refresh keeps prior data on failure
Metadata	Stored collection time, forecast issue time and source with each record so the time basis of an LLM answer can be checked
Scale & tests	~10K tourism records, 8 marine forecast feeds; 195 Python and 28 Node tests passing; Langfuse execution tracing

④ V-PASS vessel observation datasets

GeoSR · data years 2023 - 2025

Processing	Annual streaming merge, speed and vessel-ID filters, monthly/seasonal aggregation, regional grid assignment and count sums, weighted Quartic KDE, catch-grade join
Validation	Exclusion, grid-assignment and count reconciliation reports matching input and output totals; methodology and validation records public

⑤ Forecast system build, operation & analysis

GeoSR · 2023 - 2025

Build	Forecast-index and sea-condition chart production servers: quotes, install, remote access, packet filtering, Oracle DB, Tomcat, Python/Java; backup NAS (2025)
Operation	Production scheduling, daily output monitoring, incident response and reporting
Analysis	Old vs. new model product comparisons, severe-weather-day and website/OpenAPI usage analysis, monthly statistics and summaries